

EXERCISE 2 SQL FOUNDATIONS

(Aggregate Functions & Operators)

Question 1

(1) **SELECT DISTINCT** department
FROM Students ;

Students

departments
IT
HR
Finance

(2) **SELECT** department,
AVG (age) **AS** avg_age
FROM Students
GROUP BY department ;

Students

departments	avg_age
IT	20,5
HR	22
Finance	23

(3) **SELECT** department, COUNT * **AS** student_count
FROM Students
GROUP BY department
HAVING COUNT * > 1 ;

Students

department	Student_Count
IT	2
HR	2

(4) **SELECT** *
FROM Students
WHERE age **BETWEEN** 21 **AND** 23;

Students

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

(5) **SELECT** *
FROM Students
WHERE department **IN** ('IT', 'HR')
AND age > 21;

Students

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

(6) **SELECT** department,
Sum(credits) **AS** total_credits
FROM Courses
Group By department
HAVING SUM(credits) > 5;

8)

Courses

department total_credits

IT

||

(7) **SELECT** Course_id,
 Course_name,
 department,
 Credits

Course_id	Course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

FROM Courses

WHERE Credits <> 4;

(8) **SELET** course-id,
 Course-name,
 Credits

FROM Courses

ORDER BY Credits **DESC**, course-id **ASC**

LIMIT 3;

COURSES

Course id	Course name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

(9) **SELECT** MAX (grade) **AS** max_grade,
 MIN (grade) **AS** min_grade,
 AVG (grade) **AS** avg_grade

FROM enrollments;

Enrollments

max_grade	min_grade	avg_grade
90	78	84.5

10. **SELECT** course_id,
COUNT * **AS** enrollment_count
FROM enrollment
GROUP BY course_id ;

enrollments

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. **SELECT** department ,
SUM(salary) **AS** total_salary,
SUM(bonus) **AS** total_bonus
FROM Salaries
GROUP BY department ;

Salaries

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. **SELECT** department,
AVG(salary) **AS** avg_salary
FROM Salaries
GROUP BY department
HAVING Avg(salary) > 55000 ;

Salaries

department	avg-salary
IT	61000
Finance	70000

13. **SELECT** employee-id,
 name,
 department,
 salary,
 bonus, salary + bonus **AS** total-compensation
FROM Salaries
WHERE salary + bonus > 60000;

Salaries

employee-id	name	Salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500
5	Butch			

14. **SELECT** department
 SUM(budget) **AS** total-budget,
 AVG(budget) **AS** avg-budget
FROM Projects
GROUP BY department
HAVING AVG(budget) > 70000;

Projects

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

15. **SELECT** project_id,
 project_name,
 department,
 budget
FROM Projects
WHERE budget **BETWEEN** 50000 **AND** 120000
AND department $\neq$ 'Marketing';

Projects

Project_id	Project_name	department	budget
2	Payroll System	Finance	80000
4	Website	Marketing ^{Excluded}	60000
5	HR Portal	HR	50000